

James Oswald Nde Naoussi

Google DeepMind Scholar | Physics & AI for Science

☎ (+27) 068 821 7103 / (+237) 671 959 183 | ✉ naoussi.oswald@gmail.com | 📍 Muizenberg, Cape Town, South Africa

🌐 [james-naoussi](#) | 🐙 [dev-svald](#) | 🗣️ French & English (bilingual)

PROFESSIONAL PROFILE

Physicist and Google DeepMind Scholar in the AI for Science stream at the African Institute for Mathematical Sciences (AIMS), South Africa. I combine a rigorous training in mathematics and physics with a growing specialisation in machine learning, artificial intelligence and quantum computing. I am motivated by teaching, clear scientific communication, and building solutions that translate theory into measurable impact. My goal is to strengthen my expertise across mathematics, science, and the use of AI to tackle complex, real-world problems with creativity and rigour.

EDUCATION

African Institute for Mathematical Sciences (AIMS) <i>AI for Science</i> — MSc in Mathematical Sciences	Cape Town, South Africa Sept. 2025 – Present
• Recipient of the Google DeepMind Scholarship. • Focus: bridging the intersection of mathematics, machine learning and AI to address challenges in different science disciplines. • Courses included: Machine learning, Quantum Computing, NLP/LLM, Reinforcement Learning, Matrix and Tensor factorisation, Fluid Dynamics.	
University of Buea <i>MSc in Physics</i>	Buea, Cameroon Oct. 2022 – Dec. 2024
• Courses included: Quantum mechanics, Mathematical Methods, Electrodynamics, Continuum Dynamics. • GPA: 3.65	
University of Buea <i>BSc in Applied Physics</i>	Buea, Cameroon Oct. 2016 – Dec. 2019
• Courses included: Digital electronics, Communication systems, Microprocessors, Computer Architectures and Data Network. • GPA: 3.20	
Bilingual Grammar School Molyko <i>Baccalauréat en Sciences Mathématiques</i>	Buea, Cameroon Sept. 2015 – June 2016

PROFESSIONAL EXPERIENCE

Private Tutor — Mathematics & Applied Sciences <i>Self-employed; long-term client relationships across secondary and undergraduate levels</i>	Buea, Cameroon Sept. 2020 – June 2025
Bilingual Grammar School Molyko <i>Secondary School Mathematics Teacher</i>	Buea, Cameroon Oct. 2022 – June 2023

RESEARCH PROJECTS

Movie Recommender System using ALS and Matrix Factorization	Nov. 2025 (unpublished)
Automated Essay Scoring (AES): Human–AI Collaboration with Budget and Fairness Constraints	Mar. 2026 (unpublished)
Deep Learning for Quantum Error Mitigation using Classical Shadows	May. 2025 (on-going)

PUBLICATION

High intensity and frequency modulations based on infinite pulse trains and ultrashort pulses	2025 CHAOS 35 (8)
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WORKSHOPS & SHORT PROGRAMMES

Africa Scientifique <i>Skills in scientific communication</i>	AIMS South Africa 25–27 Feb. 2026
Theory of Quantum Machine Learning algorithms <i>Bridging the intersection of Quantum Computing and Machine Learning</i>	AIMS South Africa 27–30 Oct. 2025
Creative Genius Hut <i>Social & entrepreneurial skills</i>	Buea, Cameroon 23 June – 25 July 2025

LEADERSHIP, SERVICE & COMMUNITY

AIMS South Africa

Class Representative and Club President

Cape Town, South Africa

Dec. 2025 – Present

Creative Genius Hut

Team Lead

Buea, Cameroon

June – July 2025

Former Parish (Youth Ministry)

Youth Leader

Buea, Cameroon

Sept. 2020 – Aug. 2025

CORE COMPETENCIES

- **Scientific computing & ML:** Python, MATLAB, introductory machine-learning workflows, reproducible analysis.
- **Physics & mathematics:** Applied & mathematical physics training; modelling, simulation, and analytical reasoning.
- **Communication:** Classroom instruction, technical writing, science storytelling for non-specialists.
- **Digital collaboration:** Social media for outreach, remote coordination, community building.

BEYOND THE LAB

Personal interests: travelling, hiking, choral singing, reflective reading.

I recharge through nature walks and cultural exploration; music keeps me grounded creatively, while wide reading feeds curiosity across science communication, leadership, and futures thinking.

REFERENCES

Prof. Ulrich Paquet, Director of the AI for Science Master

African Institute for Mathematical Sciences (AIMS), South Africa

ulrich@aims.ac.za

Prof. Dikande Alain Moise, Head of Physics Department

University of Buea, Cameroon

dikande.alain@ubuea.cm

Prof. Eric Fokam, Head of Zoology and Animal Physiology Department

University of Buea, Cameroon

fokam.eric@ubuea.cm

Nationality: Cameroonian | Born: 13 July 2001, Buea | WhatsApp: +237 671-959-183